

Git 的運作核心在於三個主要區域 (Working Directory, Staging Area, Repository)。理解這三個階段如何互動，是掌握版本控制的關鍵。

Git 的三個核心階段原理

1. 工作目錄 (Working Directory / Workspace)

- **定義：**這就是你電腦上實際看得到的資料夾。當你編輯程式碼、新增檔案時，都是在這裡操作。
- **狀態：**處於「未追蹤 (Untracked)」或「已修改 (Modified)」的檔案會留在這裡。

2. 暫存區 (Staging Area / Index)

- **定義：**這是一個緩衝區，位於工作目錄與版本庫之間。它記錄了「下一次提交 (commit) 要包含哪些變更」。
- **原理：**Git 不會一次把所有變更提交，而是讓你精準控制內容。使用 `git add` 就是將檔案的「快照」放入這個區塊。

3. 版本庫 (Repository / HEAD)

- **定義：**這是 Git 的資料庫（`.git` 目錄）。這裡存放了完整的專案歷史紀錄，每一個 `commit` 都是一個靜態快照。
 - **原理：**當執行 `git commit` 時，Git 會將暫存區的內容永久封裝並寫入版本庫。
-

指令與階段流向詳細說明

指令	作用域	原理說明
<code>git add</code>	工作區 → 暫存區	將檔案目前的變更內容加入索引 (Index)，準備進入下一次提交。
<code>git commit</code>	暫存區 → 版本庫	將暫存區的所有內容打包，建立一個永久的版本記錄 (Commit)。
<code>git checkout</code>	版本庫 → 工作區	從指定的版本庫狀態中，將檔案「覆寫」回到工作目錄中。

深入理解 `checkout` 的機制

`checkout` 是 Git 中最容易混淆的指令之一，它負責「移動指標」或「還原檔案」：

- **切換分支**：當你執行 `git checkout <branch_name>`，Git 會更新 `HEAD` 指標，並同時將版本庫中的檔案快照「更新」到你的工作目錄。
- **還原檔案**：當你執行 `git checkout -- <file>`，它是將「暫存區」或「最後一次提交」的檔案內容，直接覆蓋到「工作目錄」，用以捨棄未新增的修改。

常見流程總結

- **開發中**：在「工作目錄」修改檔案。

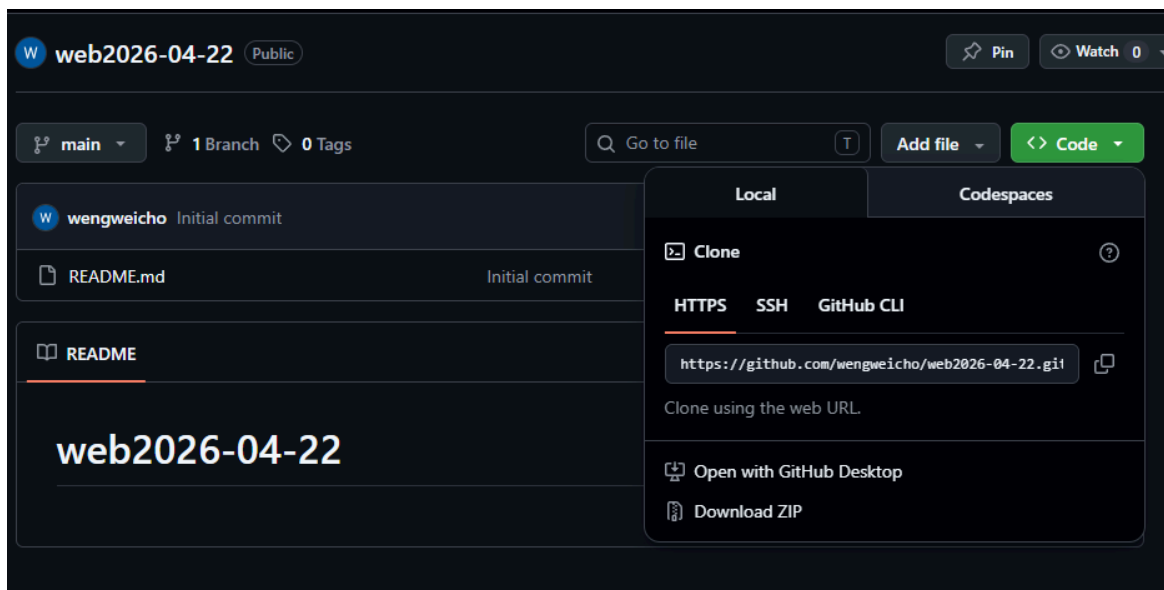
- **準備**：使用 `git add .` 將檔案移入「暫存區」。此時你可以多次 `add` 不同檔案來組裝你的提交。
- **保存**：使用 `git commit -m "描述"`，將暫存區的內容永久寫入「版本庫」。
- **回溯**：如果寫壞了，想回到某個版本，使用 `git checkout <commit_id>` 即可將版本庫內容檢出回「工作目錄」。

透過這種「先暫存、後提交」的機制，你可以確保提交內容乾淨且具備邏輯意義，避免一次送出過多不相關的變更。

=====

=====

1. 新增一個Repositories



2. git bash 切换到 d:\ & git clone

```
MINGW64:/d/obsidian/git_0422
User@1113-07 MINGW64 ~
$ cd /d/

User@1113-07 MINGW64 /d
$ cd obsidian/

User@1113-07 MINGW64 /d/obsidian
$ ls
02/      02.pdf      1150415/   '~$12.docx'
02.7z    11501-FULL-BASIC/  12.docx    '~$ple_website.docx'

User@1113-07 MINGW64 /d/obsidian
$ git clone https://github.com/wengweicho/web2026-04-22.git
Cloning into 'web2026-04-22'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

User@1113-07 MINGW64 /d/obsidian
$ ls
02/      02.pdf      1150415/   web2026-04-22/   '~$ple_website.docx'
02.7z    11501-FULL-BASIC/  12.docx    '~$12.docx'

User@1113-07 MINGW64 /d/obsidian
$ cd web2026-04-22/

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ ls
README.md

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ git remote
origin

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ git remote -v
origin https://github.com/wengweicho/web2026-04-22.git (fetch)
origin https://github.com/wengweicho/web2026-04-22.git (push)

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ cd ..

User@1113-07 MINGW64 /d/obsidian
$ git clone https://github.com/wengweicho/web2026-04-22.git git_0422
Cloning into 'git_0422'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

User@1113-07 MINGW64 /d/obsidian
$ ls
02/      02.pdf      1150415/   12.docx          '~$12.docx'
02.7z    11501-FULL-BASIC/  git_0422/  web2026-04-22/   '~$ple_website.docx'

User@1113-07 MINGW64 /d/obsidian
$ cd git_0422/

User@1113-07 MINGW64 /d/obsidian/git_0422 (main)
$ ls
README.md

User@1113-07 MINGW64 /d/obsidian/git_0422 (main)
$
```

3. git add & git commit --先更改README.md 内容

```
User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git add .

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git commit -m "web2026-04-22 version2 just change readme."
[main 84cde7e] web2026-04-22 version2 just change readme.
 1 file changed, 2 insertions(+), 1 deletion(-)

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

4. git push

```
User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ git branch
* main

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ git remote
origin

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 24 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 311 bytes | 311.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/wengweicho/web2026-04-22.git
    cec5908..84cde7e  main -> main
```

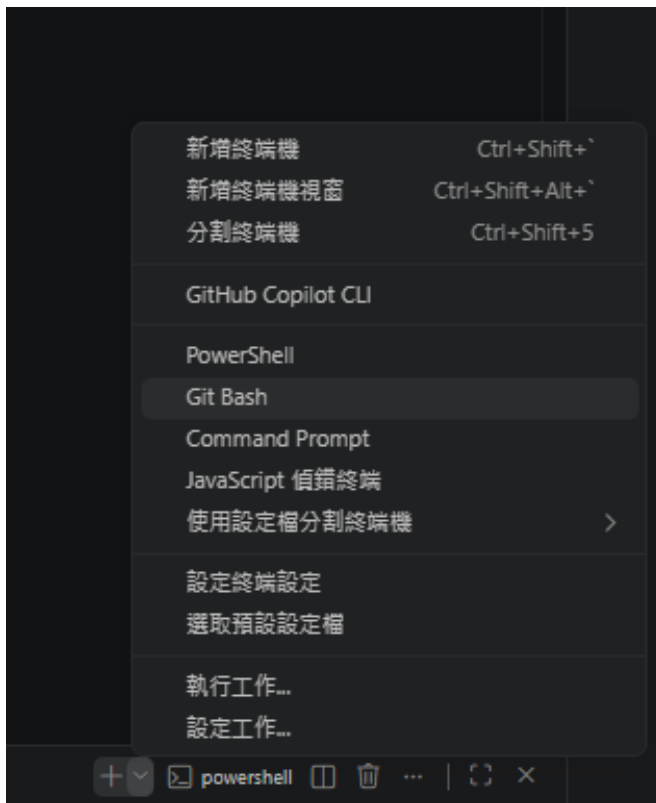
5. git remote & git remote -v

```
User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ git remote
origin

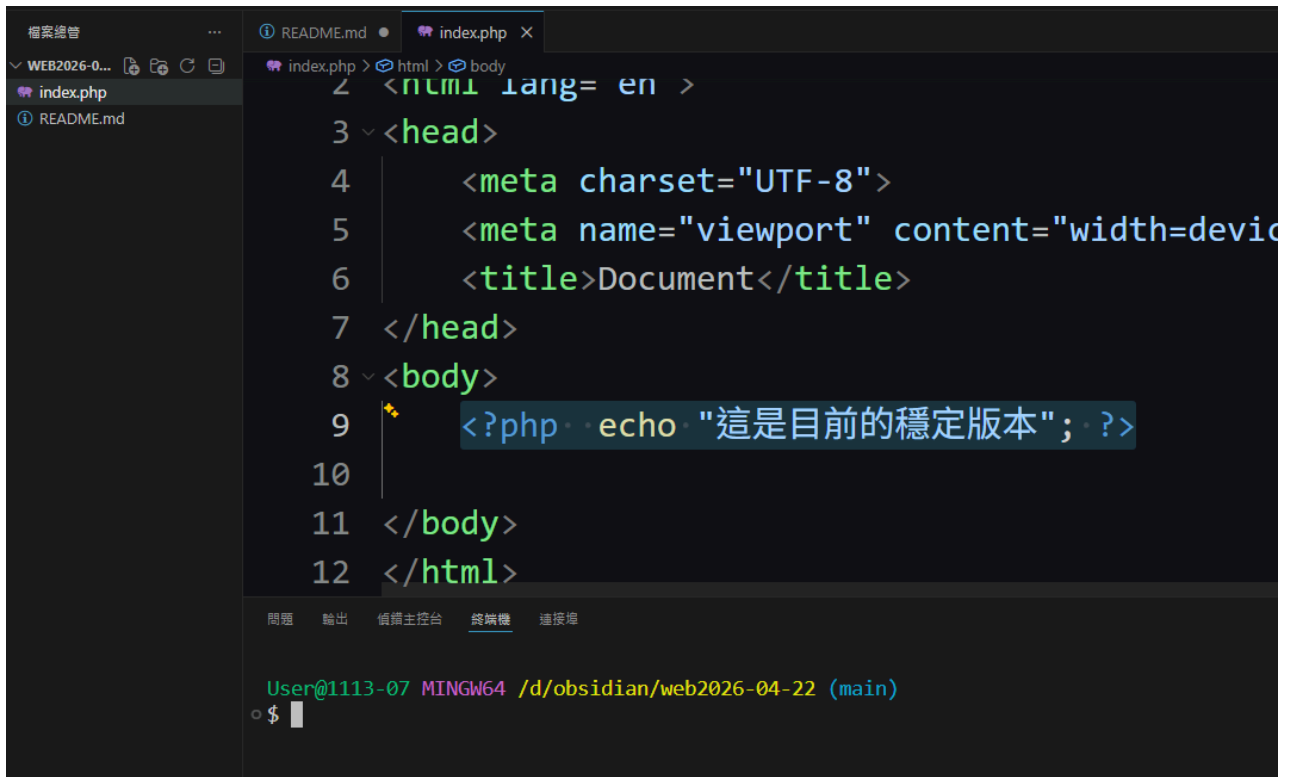
User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ git remote -v
origin https://github.com/wengweicho/web2026-04-22.git (fetch)
origin https://github.com/wengweicho/web2026-04-22.git (push)

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ |
```

6. 在下圖往下箭頭按下滑鼠右鍵,就會出現方框,可選power shell或Git Bash



7. git check out



```
User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ git status
(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
    modified:   index.php

no changes added to commit (use "git add" and/or "git commit -a")

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git checkout -- index.php

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $
```

8. 路徑問題

```
User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$ git status
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   php/test1.php

no changes added to commit (use "git add" and/or "git commit -a")

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git checkout -- "D:\obsidian\web2026-04-22\php\test1.php"

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
$
```

這樣也可以

```
User@1113-07 MINGW64 /d/obsidian/web2026-04-22 (main)
• $ git checkout -- ./php/test1.php
```


9. 更改git bash的文字大小

